

Deep Learning

Variational Autoencoder

Autoencoders

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Autoencoders

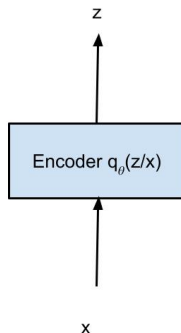
- ① Designed to reproduce input, especially reproduce the input from a learned encoding
- ② We attempted to project the data into the latent space and model it via a probability distribution
- ③ This wasn't satisfying

Variational Autoencoders

- ① 'Regularized' autoencoder to enforce latent space 'organization'

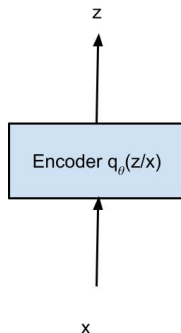
Variational Autoencoders

- ① Key idea is to make both Encoder and Decoder stochastic
 - instead of encoding an i/p as a single point, we encode it as a distribution over the latent space



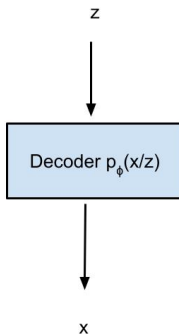
Variational Autoencoders

- ① Key idea is to make both Encoder and Decoder stochastic
 - instead of encoding an i/p as a single point, we encode it as a distribution over the latent space
- ② Latent variable z is drawn from a probability distribution for the given input x



Variational Autoencoders

- ① Then, the reconstruction is chosen probabilistically from the sampled z



VAE Encoder

- ① Takes i/p and returns the parameters of a probability density (e.g. Gaussian, mean and covariance matrix)

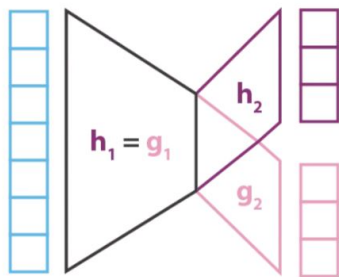
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- ③ NN implementation of the encoder gives (for every input x) a vector mean and a diagonal covariance

VAE Encoder



x

$$\mu_x = g(x) = g_2(g_1(x))$$

$$\sigma_x = h(x) = h_2(h_1(x))$$

VAE Decoder

- 1 Decoder takes the latent vector z and returns the parameters for a distribution

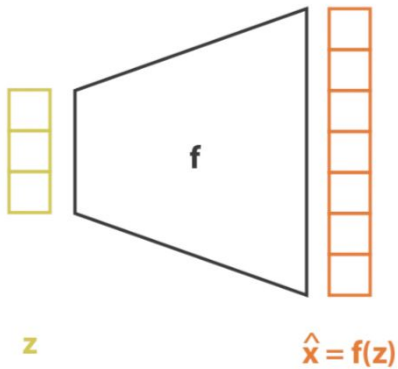
VAE Decoder

- ① Decoder takes the latent vector z and returns the parameters for a distribution
- ② $p_{\phi}(x/z)$ gives mean and variance for each pixel in the output

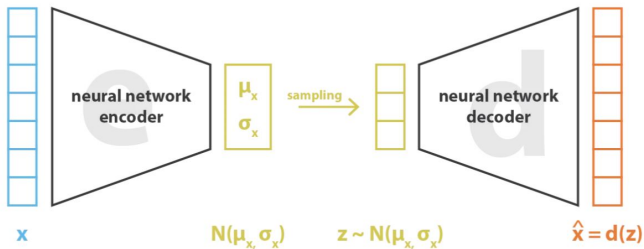
VAE Decoder

- ① Decoder takes the latent vector z and returns the parameters for a distribution
- ② $p_{\phi}(x/z)$ gives mean and variance for each pixel in the output
- ③ Reconstruction of x is via sampling

VAE Decoder



VAE Forward pass



VAE loss function

- ① Loss for AE: l_2 distance between the input and its reconstruction

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- ② In case of VAE: we need to learn parameters of two probability distributions
- ③ For each input x_i we maximize expected value of returning x_i (or, minimize the NLL)

$$-\mathbb{E}_{z \sim q_{\theta}(z/x_i)}[\log p_{\phi}(x_i/z)]$$

VAE loss function

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- ① Problem: Input images may be memorized in the latent space
- $\rightarrow$ similar inputs may get different representations in z space
 - $\rightarrow$ close points in the latent space should not give two completely different contents once decoded

VAE loss function

$$-\mathbb{E}_{z \sim q_{\theta}(z/x_i)} [\log p_{\phi}(x_i/z)]$$

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- ① Continuity and Completeness: We prefer continuous latent representations to give meaningful parameterization (e.g. smooth transition between i/ps)
- ② Solution: Force $q_{\theta}(z/x_i)$ to be close to a standard distribution (e.g. Gaussian)

VAE loss function

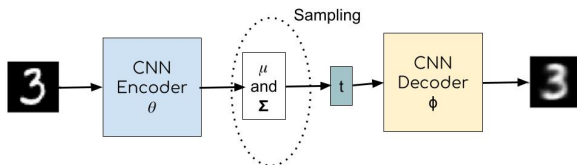
$$l_i(\theta, \phi) = -\mathbb{E}_{z \sim q_\theta(z/x_i)}[\log p_\phi(x_i/z)] + \mathbb{KL}(q_\theta(z/x_i) || p(z))$$

- ① First term promotes recovery, sencond term keeps encoding continuous (beats memorization)

VAE loss function

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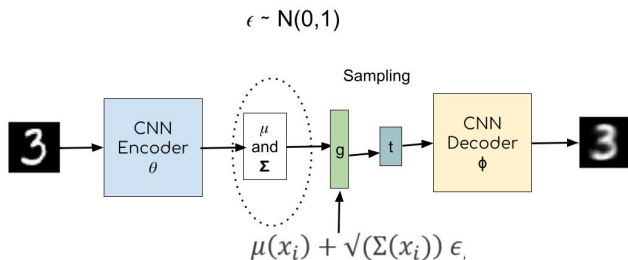
① Problem: Differentiating over θ and ϕ



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- ① Reparameterization: Draw samples from $N(0,1) \rightarrow$ doesn't depend on parameters



Generation with VAE

- Sample z from the prior $p(z)$

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- For simplicity, in practice, only the means of the pixels are inferred (deterministic)

Generation with VAE



Figure credits: Wojceich

Generation with VAE

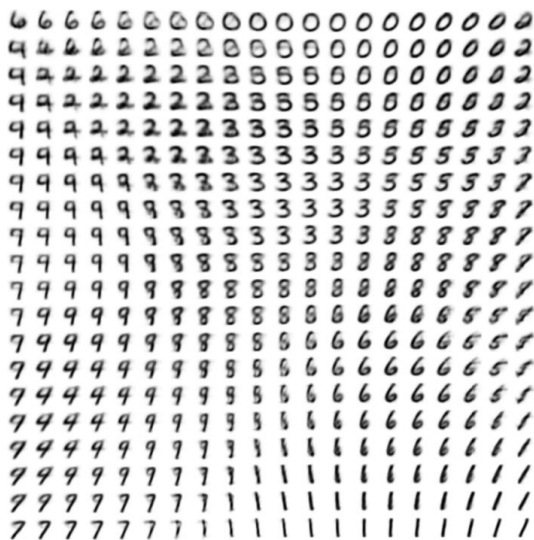


Figure credits: Kingma et al.

Edit/Manipulate samples with VAE



The Evidence Lower Bound (ELBO)

Latent Variable Models

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 - account for measurement errors
 - controlled/customized generation of the samples

Latent Variable Models

- ① They model the probability distribution over latent variables

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- ② Because the latent variables explain the data in a simpler way

Latent Variable Models - terminology

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- ③ $p(z)$ prior distribution that models the behavior of latent variables

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- ③ Marginal distribution $p(x)$ (goal of the model) describes how likely a sample is
- ④ $p(z/x)$, posterior, describes the latent variables that can be produced by a data sample

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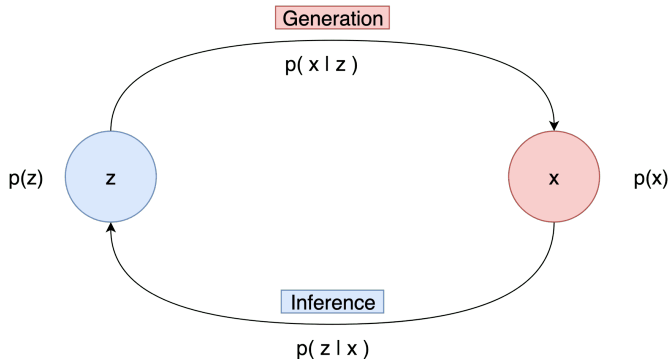
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Latent Variable Models - terminology

- ① Inference - process of finding the latent variable z from the data point x
- ② Formulated by the posterior distribution $p(z/x)$

Generation-Inference

- If we assume that we (somehow) know the likelihood $p(x/z)$, the posterior $p(z/x)$, the marginal $p(x)$, and the prior $p(z)$



Aim/Question of latent Variable Models

- ① How to find these distributions?

These can be connected

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is intractable $\rightarrow$ can't compute the LHS

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⑤ Variational inference suggests to use another (known) distribution $q_{\phi}(z/x)$ to approximate the posterior $\rightarrow$ (allows to compute the evidence and sample)

Variational Inference

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- ④ KL Divergence

Variational Inference

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- ③ $D_{KL}(q_\phi || p_\theta) = \mathbb{E}_{q_\phi} [\log q_\phi(z/x)] - \mathbb{E}_{q_\phi} [\log p_\theta(z/x)]$

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- ④ We can't compute because we don't have its analytical form

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- ⑤ This is the lower bound on the evidence
- ⑥ Now, in order to reduce the D_{KL} , we can maximize the ELBO

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- ⑥ VAEs model $p_\theta(x/z)$ and $q_\phi(z/x)$ as neural networks